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A DETAILS ON THE ANNOTATION PROCESS

We use the following prompt to automatically annotate LLM responses:

```
Please split the following reasoning chain of an LLM into
annotated parts using labels and the following format ["label
"]...["end-section"]. A sentence should be split into multiple
parts if it incorporates multiple behaviours indicated by the
labels.

Available labels:
0. initializing -> The model is rephrasing the given task and
states initial thoughts.
1. deduction -> The model is performing a deduction step based on
its current approach and assumptions.
2. adding-knowledge -> The model is enriching the current approach
 with recalled facts.
3. example-testing -> The model generates examples to test its
current approach.
4. uncertainty-estimation -> The model is stating its own
uncertainty.
5. backtracking -> The model decides to change its approach.

The reasoning chain to analyze:
{thinking_process}

Answer only with the annotated text. Only use the labels outlined
above. If there is a tail that has no annotation leave it out.
```

B ANNOTATED EXAMPLE

Riddle: What has cities, but no houses; forests, but no trees; and rivers, but no water?

DeepSeek R1 Response:

```
Okay, so I came across this riddle. At first glance, it seems
tricky, but I can break it down.
Let me analyze each part separately. The first part: "What has
cities, but no houses." The answer must be something that contains
 cities but lacks houses. Cities have buildings, roads, and
infrastructure, but without houses, it must be something that
represents a city rather than an actual one.
Maybe it's a city without residential buildings, but that doesn't
fit perfectly.
Wait, perhaps it refers to something that *displays* cities rather
 than being a real city.
What about the second part: "forests, but no trees"? Forests are
made of trees, so if there are no trees, it isn't really a forest.
 But perhaps it's something that symbolizes a forest without
actually being one.
Finally, "rivers, but no water." Rivers are defined by flowing
water, so if there's no water, it might be referring to a
representation of a river rather than a real one.
Considering all three parts together, the answer must be something
 that includes symbolic representations of cities, forests, and
rivers but lacks their real-world components. A map fits this
description-it has cities, forests, and rivers marked on it, but
they are not real.
```